

1. Signatures characterization:

The constructors are:

zero	:	()	-->	poly
make-poly	:	Number X Number	-->	poly
add-poly	:	poly X poly	-->	poly

The observers are:

degree	:	poly	-->	Number U {error: degree zero poly}
Coeff	:	poly X Number	-->	Number
print-poly	:	poly	-->	List
Calc-poly	:	poly X Number	-->	Number

The predicates are:

is-zero?	:	poly	-->	Boolean
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2. Algebraic characterization:

The constructors are:

$$(\text{zero}) = 0$$

$$(\text{make-poly coefficient exponent}) = [f]$$

Where $[f] = \text{coefficient} \cdot x^{\text{exponent}}$

$$(\text{add-poly } [f] [g]) = [h]$$

Where $[h] = [f] + [g]$

The observers are:

$$(\text{degree } [f]) = n \quad (n \geq 0)$$

Where $n = \max\{\max \text{all exponents in } [f], 0\}$
And $[f] \neq 0$

$$(\text{coeff } [f] n) = a$$

Where $a = \begin{cases} c & , \text{ if } c \cdot x^n \text{ is a term in } [f] \\ 0 & , \text{ else} \end{cases}$

$$(\text{print-poly } [f]) = [g]$$

Where $[g] = \begin{pmatrix} (\text{coefficient of first term of } [f], \text{exponent of first term of } [f]), \\ (\text{coefficient of second term of } [f], \text{exponent of second term of } [f]) \\ \vdots \\ \vdots \end{pmatrix}$

$$(\text{Calc-poly } [f] x) = m$$

Where $m = \sum_{i=1}^{\text{amount of terms in } [f]} c_i \cdot x^{e_i}$
And c_i is the i 'th coefficient in $[f]$
And e_i is the i 'th exponent in $[f]$

The predicates are:

$$(\text{is-zero? } [f]) = \begin{cases} \#t & , \text{ if } [f] = 0 \\ \#f & , \text{ if } [f] \neq 0 \end{cases}$$

Special occasions:

(degree $[\emptyset]$)

= *error*